

Experiment 1: Case Study

Problem Statement: Choose the KL ERP System case study, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

1. Functional Requirements

Faculty

- Record attendance
- Upload course materials
- Enter student marks
- Manage class schedules
- Communicate with students

Administration

- Manage student records
- Manage faculty records
- Manage course allocations
- Generate academic reports
- Monitor institutional data

2. User Stories

User ID	User Story	Priority
US-01	As a faculty member, I want to record attendance so that student attendance is tracked accurately.	High
US-02	As a faculty member, I want to upload course materials so that students can access learning resources.	High
US-03	As a faculty member, I want to enter marks so that students can view	High

	their academic performance.	
US-04	As an administrator, I want to manage student records so that all student information remains updated.	High
US-05	As an administrator, I want to generate reports so that institutional performance can be monitored.	Medium
US-06	As an administrator, I want to manage faculty and course allocations so that academic activities are organized efficiently.	Medium

3. Selected Software Development Process Model

AGILE SCRUM

Reason for Selection

- Requirements change frequently.
- Continuous feedback from students, faculty, and administrators.
- Incremental feature delivery.
- Faster development cycles.
- Easy bug fixing.
- Suitable for ERP applications.

Sprint Plan

Sprint	Features
Sprint 1	Faculty Login & Attendance
Sprint 2	Course Materials
Sprint 3	Marks Module
Sprint 4	Student Records
Sprint 5	Reports
Sprint 6	Testing & Deployment

4. Programming Language Selection

Backend

Java (Spring Boot)

Reasons

- Secure
- Scalable
- Enterprise-level applications
- REST API support
- Excellent database connectivity

Frontend

- HTML5
- CSS3
- JavaScript
- React.js

Database

MySQL

Other Technologies

- REST API
- JWT Authentication
- Git
- Docker
- AWS Cloud

5. System Architecture

Faculty/Admin → Web Interface → Spring Boot Backend → MySQL Database

6. Implementation Strategy

Phase 1 – Requirement Analysis

- Gather requirements.
- Create SRS.
- Prepare use case diagrams.

Phase 2 – System Design

- Design database schema.
- Design UI wireframes.

- Create UML diagrams.
- Develop system architecture.

Phase 3 – Development

- Develop faculty module.
- Develop administration module.
- Integrate database.

Phase 4 – Testing

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)

Phase 5 – Deployment

- Docker
- AWS EC2
- MySQL Database
- Nginx Web Server

7. Evaluation Strategy

Evaluation Parameter	Method
Functional Accuracy	Requirement verification
Performance	Load testing
Security	Authentication testing
Reliability	Continuous monitoring
Usability	User feedback
Scalability	Concurrent user testing
Maintainability	Code review and documentation

8. Expected Outcomes

- Centralized academic management.
- Accurate attendance and marks.
- Improved reporting.
- Reduced paperwork.
- Secure and scalable ERP platform.